

Yichen Di

Tsinghua University

Email: diyc22@mails.tsinghua.edu.cn, diyc041018@gmail.com

Personal Website

Education

Tsinghua University, Beijing, China

Sept. 2022 — Present

Bachelor of Engineering in Software Engineering

GPA: 3.91/4.00, **Rank:** 3/79

Core Courses : Linear Algebra (A+), Complex Analysis (A), Discrete Mathematics (A), Introduction to Artificial Intelligence (A), Object-Oriented Programming (A), Probability and Statistics (A-), Data Structure (A-), Computer Networks (A-), Operating System (A-), Principles of Database Systems (A-), Optimization for Cloud Services (A-)

Language: TOEFL: 107, GRE: 327+3.5

Research Interests

- **AI for Science:** Developing spectral and graph-based approaches to capture physical structures in complex domains. Designing neural operators and scalable parallel methods to solve PDEs on industrial-scale geometries.
- **Large Language Models:** Studying robustness and safety of LLMs, with a focus on watermarking, adversarial attacks, and trustworthiness.

Research & Projects Experiences

Caltech Summer Research (VURP) - Learning Flow-Physics on Complex Geometries with Neural Operators

California Institute of Technology, Pasadena, CA

May 2025 — Present

Advisor: Prof. Anima Anandkumar

- **Architecture Refinements and Model Size Tuning:** Proposed architecture refinements to FIGConv, such as replacing local convolution with interpolation, for accuracy and efficiency. Also reduced the model size to make it more generalizable.
- **Geometric Conditioning:** Explored adding different physical parameters (e.g., inflow velocity, rotational speed) and signed distance function (SDF) embeddings with alternative integration methods to assess their effect on accuracy.
- **Generalization Gap:** Eemonstrated strong overfitting capacity ($\sim 0.1\%$) but persistent $\sim 28\text{--}30\%$ error on unseen data, highlighting data scarcity as the limiting factor.

Transolver++: An Accurate Neural Solver for PDEs on Million-Scale Geometries

Tsinghua University, Beijing, China

Sept. 2024 — Mar. 2025

Advisor: Prof. Mingsheng Long

- **Adaptive Mechanism Design:** Designed and compared single-scaling and joint-scaling attention strategies, significantly improving Transolver++'s ability to capture fine-grained local features in PDE solutions.
- **Scalability Enhancement:** Identified per-point mapping as the key bottleneck and introduced data parallelism, extending the solver's scalability from million-level to tens-of-million-level meshes.
- **Slice Analysis:** Conducted KL-divergence analysis of slice weight distributions to demonstrate that Transolver++ achieves more informative representations ($3.885 \rightarrow 5.297$), explaining its superior performance over Transolver.

WaterSeeker: Pioneering Efficient Detection of Watermarked Segments in Large Documents

Tsinghua University, Beijing, China

May 2024 — Aug. 2024

Advisor: Prof. Lijie Wen

- **Formulation of Detection Metrics:** Derived and implemented z-score formulations for KGW and AAR watermarks, enabling accurate suspicious region detection within the unified WaterSeeker framework.
- **Algorithmic Refinement:** Introduced a tolerance coefficient into segment concatenation to address incomplete region detection, improving robustness in watermark localization.
- **Comparative Validation:** Compared WaterSeeker with WinMax and FLSW variants on KGW, showing that WaterSeeker achieves $F1 = 0.872$, substantially outperforming FLSW ($F1 \leq 0.822$) under comparable efficiency.

MarkLLM: An Open-Source Toolkit for LLM Watermarking (GitHub, 620 stars)

Contributor of Algorithms and pipelines for testing and visualization

May 2024 — July 2024

Advisor: Prof. Lijie Wen

- **Algorithm Implementation & Evaluation Pipeline:** Implemented multiple watermarking methods (KGW, Unigram, SWEET) within MarkLLM’s unified framework, and integrated them into automated pipelines covering detectability, robustness, and text quality.
- **Mechanism Visualization Design:** Built an extensible visualization module that highlights token-level watermarking mechanisms for both KGW and Christ families. Designed tailored rendering strategies and configurable options for users.

Document-Level Relation Extraction with Collaboration of Large and Small Models

Principle Investigator

Nov. 2023 — May 2024

Advisor: Prof. Lijie Wen

- **Prompt Design & Model Enhancement:** Designed prompts to guide language models in extracting entities and judging relationship predictions and attempted improvements to the DREEM model for entity-pair relation extraction.
- **Adaptive Evaluation Strategy:** Proposed a logit-based allocation mechanism that dynamically assigns samples from the small model to the large one, enhancing efficiency while preserving extraction accuracy.

Corner Case Analysis and Benchmarking of Large Language Models

Tsinghua University, Beijing, China

Oct. 2022 — Aug. 2023

Advisor: Prof. Jie Tang

- **Data Collection & Categorization:** Collected reasoning and commonsense failure cases from online sources and organized them into structured benchmark sets.
- **Prompt Attack Exploration:** Designed adversarial prompts and jailbreak strategies to analyze refusal bypass and vulnerability patterns in ChatGPT.

Publications

- **Transolver++: An Accurate Neural Solver for PDEs on Million-Scale Geometries** (arXiv)
Huakun Luo, Haixu Wu, Hang Zhou, Lanxiang Xing, **Yichen Di**, Jianmin Wang, Mingsheng Long
ICML 2025
- **WaterSeeker: Pioneering Efficient Detection of Watermarked Segments in Large Documents** (arXiv)
Leyi Pan, Aiwei Liu, Yijian Lu, Zitian Gao, **Yichen Di**, Lijie Wen, Irwin King, Philip S. Yu
NAACL 2025 Findings, AAAI 2025 PDLM Oral

Technical Skills

- **Programming:** Python, Pytorch, C++, Java, Verilog
- **Frameworks:** Transformer, UNet, CNN, GNN, Neural Operators, Diffusion Models
- **Software:** Git, Linux, Visual Studio Code, IntelliJ IDEA, Overleaf

Extracurricular Experiences

Tsinghua Global Youth Dialog (TGYD)

Key Member, Events Department

Oct. 2023 — Present

As a key member of the Events Department, I organized “Roundtable Forums” with entrepreneurs and experts, coordinated cross-functional student teams, and assisted in guest reception and logistics to ensure smooth communication. I also engaged in cross-cultural collaboration by liaising with overseas universities and companies, facilitating international partnerships and practical activities.

Student Activities

Tsinghua University

Sept. 2022 — Present

I actively participated in diverse student activities, representing my department in campus singing competitions and collaborating with peers to deliver public performances. In addition, I was a dedicated member of the college football and frisbee teams, where I developed teamwork, resilience, and strategic communication through competitive practice.